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SHORT COMMUNICATION

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Human Herpesvirus 8 seroprevalence among blood donors in Mali

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Abstract

In sub-Saharan Africa, the Human Herpesvirus 8 (HHV-8) is endemic but with disparities between regions and population studied. Although the virus remains mostly latent, there is some evidence that blood transfusion may represents one of the transmission way for this virus. Here, we evaluated HHV-8 seroprevalence among blood donors in Mali. This cross-sectional study recruited blood donors from the Blood Transfusion Center at Gabriel Touré Hospital, Bamako. Serum was used for the detection of latent HHV-8 immunoglobulin G directed against latent associated nuclear antigen 1 by an indirect immunofluorescence assay. Human immunodeficiency virus 1 (HIV-1), Hepatitis B Virus (HBV), HCV, and *Treponema pallidum* were also screened. HHV-8 seroprevalence was 10.4% in Malian blood donors. None of the sociodemographic characteristics were associated with HHV-8 infection, although there is a tendency of a higher HHV-8 seroprevalence among participants living in Bamako than those not living there. One individual had coinfection HHV-8/HBV, another HHV-8/HCV while another had HCV and *T. pallidum*. None has been tested positive for HIV infection. This intermediate seroprevalence in Malian blood donors suggests that the risk of HHV-8 transmission by transfusion should be considered. Further investigations are needed to assess impact of HHV-8 in polytransfused patients residing in an endemic area for this virus.

KEYWORDS

Blood donors, HHV-8, IFA, Mali, Seroprevalence, Transfusion

Abbreviations: HHV-8:, Human Herpesvirus 8; IFA, immunofluorescence assay; KSHV, Kaposi's sarcoma-associated herpesvirus; LANA-1, latent associated nuclear antigen 1.

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1 | INTRODUCTION

Human herpesvirus 8 (HHV-8), also known as Kaposi's sarcoma-associated herpesvirus (KSHV), causes several neoplastic diseases and is responsible of all forms of KS. HHV-8 infection is not ubiquitous, but endemic in sub-Saharan Africa, some parts of Eastern Europe and Mediterranean. HHV-8 seroprevalence varies between different populations and it is commonly found in human immunodeficiency virus (HIV)-positive individuals and men who have sex with men.[1,2] The exact route of transmission is still very much debated. Some studies report that saliva is implicated as the main vehicle of transmission in sub-Saharan children; others, on the other hand, support the hypothesis of horizontal transmission in adulthood.[2,3] It should be noted that, HHV-8 can be transmitted by blood transfusion. Ensuring good haemovigilance remains a major public health problem in sub-Saharan Africa and more particularly in low-income countries on the African continent.[4] Various studies provide strong evidence of HHV-8 transmission by blood transfusion in sub-Saharan Africa. An Ugandan study carried out on patients receiving transfusion; tested for anti-HHV-8 antibodies pre- and post-transfusion, showed that 43% received blood seropositive for HHV-8 and seroconversion risk was significantly higher in recipients of HHV-8 seropositive blood.[5] In addition, this area is described as highly endemic for this virus. Some subregion countries reported HHV-8 seroprevalences in blood donors of 14%, 22%, and 57% in Burkina Faso, Central African Republic and Tanzania, respectively.[2,6,7] In Mali, HHV-8 studies in blood donors are almost absent while the most dreaded posttransfusion infections are viral in origin. We investigated the prevalence of antibodies directed against HHV-8 latent associated nuclear antigen 1 (LANA-1) by use of an indirect immunofluorescence assay (IFA) in blood donors from Bamako and its periphery.

2 | MATERIALS AND METHODS

This cross-sectional study included 229 parental and voluntary nonremunerated blood donors, without clinical evidence of Kaposi Sarcoma recruited from November 2019 to January 2020 at the Blood Transfusion Center at Gabriel Touré Hospital in Bamako, Mali. After collecting sociodemographic characteristics, serum was collected for HIV-1, hepatitis B surface antigen (HBsAg), hepatitis C virus (HCV), and syphilis screening in Bamako blood bank laboratory. The detection of HHV-8 antibodies directed against LANA-1 immunoglobulin G by IFA using the BC-3 cell line infected with HHV-8 but not with Epstein–Barr virus (Figure 1 and Supporting Information: Figure 1) was performed as previously described in the virology department of the Pitié-Salpêtrière Hospital, Paris, France.[8] This technique uses unstimulated cells (sensitivity 80%–85%, specificity nearly 100%). Samples reactive at an 1:50 dilution were

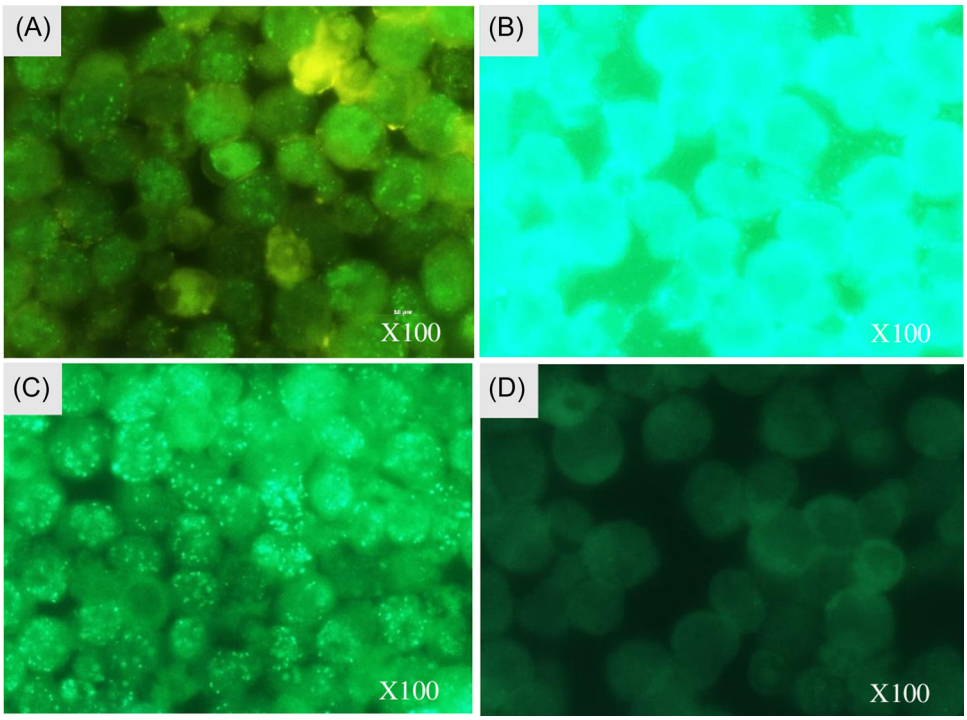


FIGURE 1 Indirect Immunofluorescence assay for the qualitative detection of HHV-8 IgG antibodies directed against the LANA-1 protein. The positivity for HHV-8 antibodies is revealed by the presence of dots in the nucleus. (A) Positive sample at 1:50 dilution; (B) indeterminate result sample at 1:50 dilution (equivocal HHV-8 serology): ambiguous result, hyperfluorescence on reading; (C) strongly positive sample at the 1:50 dilution; (D) negative sample at 1:50 dilution. HHV-8, Human Herpesvirus 8; IgG, immunoglobulin G; LANA-1, latent associated nuclear antigen 1.

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TABLE 1 Sociodemographic and medical data according to HHV-8 serological status in the 229 blood donors recruited.

Characteristics	Total no. (%)	HHV-8 seropositive		HHV-8 seronegative		*p* Value
		No.	%	No.	%	
All	229	23	10.4	198	89.6	
Age, median (IQR) (years)	29 (23–35)	32 (25–37)		28 (23–35)		0.24
Age group (years)						
18–30	138 (60.3)	11	47.8	124	62.6	0.25
31–40	60 (26.2)	9	39.1	50	25.3	
41–50	28 (12.2)	2	8.7	22	11.1	
>50	3 (1.3)	1	4.4	2	1	
Gender						
Male	133 (58.1)	12	52.2	115	58.1	0.65
Female	96 (41.9)	11	47.8	83	41.9	
Education level						
None	99 (43.2)	8	34.8	84	42.4	0.47
Primary	60 (26.2)	6	26.1	54	27.3	
Secondary	42 (18.3)	5	21.7	36	18.2	
Higher	28 (12.2)	4	17.4	24	12.1	
Marital status						
Single	79 (34.5)	8	34.8	70	35.4	>0.99
Married	150 (65.5)	15	65.2	128	64.6	
Geographical setting						
Bamako	203 (88.6)	18	78.3	179	90.4	0.08
Around Bamako	26 (11.4)	5	21.7	19	9.6	
Type of blood donation						
Parental donors	226 (98.7)	23	100	195	98.5	>0.99
Volunteers donors	3 (1.3)	0	0	3	1.5	
HIV-1 status						
Positive	0	0	0	0	0	NC
Negative	229 (100)	23	100	198	100	
HBsAg status						
Positive	22 (9.6)	1	4.4	20	10.1	0.70
Negative	207 (90.4)	22	95.6	178	89.9	
HCV status						
Positive	6 (2.6)	1	4.4	4	2	0.42
Negative	223 (97.4)	22	95.6	194	98	
Treponema pallidum status						
Positive	5 (2.2)	0	0	4	2.02	>0.99
Negative	216 (94.3)	22	95.6	187	94.4	
Not specified	8 (3.5)	1	4.4	7	3.6	

Abbreviations: HBsAg, hepatitis B surface antigen; HCV, hepatitis C virus; HIV, human immunodeficiency virus 1; IQR, interquartile range; NC, noncalculated.

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considered positive (Figure 1A,C and Supporting Information: Figure 1A–C,F). Note that, the term "equivocal HHV-8 serology result" was used when after twice tests in some patients, we had an indeterminate result, that is, ambiguous, fluorescent slide on reading (Figure 1B and Supporting Information: Figure 1D). Continuous variables were described with median and interquartile range (IQR) and categorical variables as numbers and percentages. GraphPad software was used to perform nonparametric tests, Mann–Whitney *U*-test for quantitative data, Fisher's exact *t*-test for qualitative data. Confidence interval was 95% (95% CI) and $p < 0.05$ was considered significant.

3 | RESULTS AND DISCUSSION

We showed that 10.4% of this population of apparently healthy blood donors in Mali had HHV-8 antibodies. None of them had clinical evidence of KS. The sociodemographic characteristics, the geographical setting and HIV-1, HBsAg, HCV and syphilis (*Treponema pallidum*) screening of the 229 blood donors are described in Table 1. The median age was 29 (IQR: 23–35) years. The majority of blood donors were male (58%), residing in Bamako (89%), attending the center for a parental blood donation (99%), and more than two thirds have reached primary school or less (69%). None of the sociodemographic characteristics tested were associated with HHV-8 infection, although there is a tendency of a higher HHV-8 seroprevalence among participants living in Bamako than those not living there. One individual had coinfection with HHV-8 and Hepatitis B Virus, another with HHV-8 and HCV while another had one with HCV and *T. pallidum*. None has been tested positive for HIV infection. Consistent with previous studies in Burkina Faso, Central African Republic and Tanzania, this blood donor population is young and predominantly male.[2,6,7] The 8 (4%) blood donors with equivocal HHV-8 serology results were excluded in our seroprevalence estimation. This is the first study to investigate HHV-8 seroprevalence in blood donors in Mali showing a moderate seroprevalence around 10%. Many samples had negative HHV-8 serology (Figure 1D and Supporting Information: Figure 1E). This seroprevalence is relatively close to that has been reported in blood donors population in Burkina Faso.[7] However, other studies carried out in blood donors from the subregion report high prevalence of up to 57%.[2,6] Differences in assays used, ethnic groups of the sampled populations, cultural practices (premasticating baby food, use of medicinal herbs), environmental factors (volcanic soil, exposure to iron), and geographic location of these countries in the area described as the KS belt could be the reason for observed differences in HHV-8 seroprevalence.[9–11] However, since the majority of transfusions are whole blood, the presence of HHV-8 in the peripheral compartment does not exclude its transmission, even in the absence of clinical symptoms. Especially since the Malian recommendations in terms of transfusion safety do not prescribe leukodepletion steps while HHV-8 is mostly latent in B lymphocytes.

Our study has some limitations: (i) the relative low sensitivity of the indirect immunofluorescence test used leads to a possible underestimation of seroprevalence in our population, (ii) the patients included in our study were all asymptomatic to KS, thus the comparison of fluorescence intensity of serology, and in view of the HHV-8 quantification in peripheral compartment, between symptomatic and asymptomatic patients could not be study, (iii) finally, the sample size studied is limited and probably explains why the statistical results do not have rich significance.

In conclusion, this intermediate seroprevalence among blood donors in Mali suggests that the risk of transmission of HHV-8 by transfusion should be considered. Further studies are needed on the evaluation of HHV-8 transmission via blood transfusion in Africa, especially in children with homozygous sickle cell disease, who generally require several transfusions, and who live in an area of high HHV-8 seroprevalence.

AUTHOR CONTRIBUTIONS

Aude Jary, Almoustapha I. Maiga, Vincent Calvez, and Anne-Geneviève Marcelin designed and planned the study. Sidi Dienta, Fatoumata Tata Traore, Zalihatou Maiga, Alhassane Ba, Ousmane Faye, and Almoustapha I. Maiga included the study patients and acquired the data. Elodie Chicaud, Stéphane Marot, and Gervillien Arnold Malonga conducted the experiments. Gervillien Arnold Malonga and Aude Jary analyzed and interpreted the data. Gervillien Arnold Malonga, Aude Jary, and Anne-Geneviève Marcelin wrote the manuscript. Gervillien Arnold Malonga, Aude Jary, Almoustapha I. Maiga, Vincent Calvez, and Anne-Geneviève Marcelin revised the manuscript. All the authors read and approved the final manuscript.

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CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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